Naïve Bayes

Fatemeh Mansoori

What Is a Random Variable?

Example: Tossing Two Coins

Let X be the number of heads. The possible values are $X \in \{0, 1, 2\}$.

Definition

A random variable is a function that assigns a numerical value to each outcome of a random experiment.

Notation

We usually use an uppercase letter (X) for a random variable and a lowercase letter (x) for a possible value or observation.

Why Do We Need It?

It allows us to describe uncertainty quantitatively using numerical values and probability distributions.

Probability Distributions

Continuous: Probability Density Function (PDF)

$f(x) \geq 0$ and $\int f(x)\,dx = 1$

The probability of an interval is the integral of the density over that interval.

Discrete: Probability Mass Function (PMF)

$p(x) = P(X = x)$

Conditions: $p(x) \geq 0$ and $\Sigma_x\, p(x) = 1$

Important Note

For a continuous variable, $f(x)$ is not the probability at x. Instead, $P(a \leq X \leq b) = \int_a^b f(x)\,dx$.

Example: A Fair Die

For $X \in \{1, 2, 3, 4, 5, 6\}$, $P(X = x) = 1/6$.

Joint and Marginal Distributions

Marginal Distribution

Describes the distribution of one variable regardless of the value of the other.

$P(X = x) = \Sigma_Y P(X = x, Y = y)$

Joint Distribution

Describes the probabilities of combinations of values of two or more random variables.

For discrete variables: $p(x, y) = P(X = x, Y = y)$

Example: X = attendance (present/absent), Y = pass/fail. The joint distribution describes each combination; the marginal distribution of X gives only the probabilities of being present or absent.

Numerical Example: From Joint to Marginal

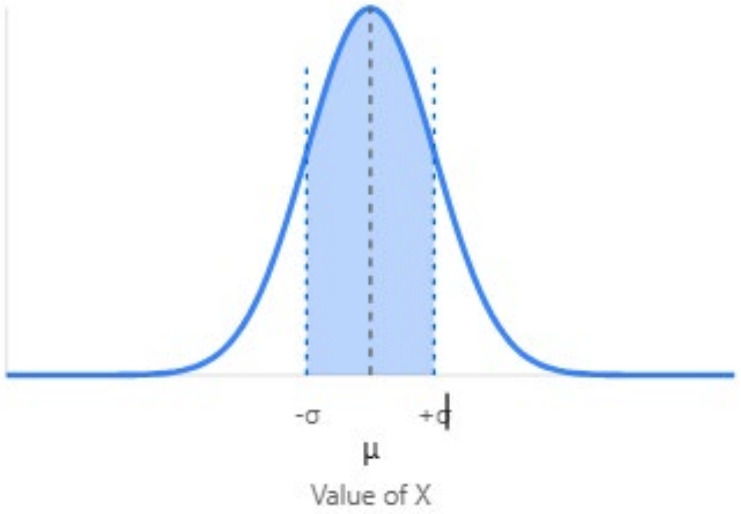
X \ Y	Y = Pass	Y = Fail	Marginal of X
X = Present	0.42	0.18	**0.60**
X = Absent	0.08	0.32	**0.40**
Marginal of Y	**0.50**	**0.50**	**1.00**

For example: P(X = Present) = 0.42 + 0.18 = 0.60

The sum of all joint probabilities is 1. Marginals are obtained by summing rows or columns.

Expectation Value and Variance

- $E[X] = \sum_x xp(X = x)$
- $E[X] = \int_{-\infty}^{\infty} xf(x)dx$



- $Var[X] = E[(X - E[X])^2]$ or $\mathrm{Var[X]} = \mathrm{E}[X^2] - E[X]^2$
- $Var[X] = \int_{-\infty}^{\infty}(x - \mu)^2 f(x)dx$

Conditional Probability

- X and Y are two random variables
- $p(X|Y) = \frac{P(X=x,Y=y)}{P(Y=y)} = \frac{P(X=x,Y=y)}{\sum_x P(X=x,Y=y)}$
- p(X,Y) = p(X|Y)p(Y)
- If X and Y are independence : p(X,Y) = p(X)p(Y)
- Bayes Theorem : $p(Y|X) = \frac{p(X|Y)P(Y)}{\sum_y p(X|Y)P(Y)}$

Definition

Posterior probability: $p(\mathcal{C}_k|\boldsymbol{x})$

Likelihood or class conditional probability: $p(\boldsymbol{x}|\mathcal{C}_k)$

Prior probability: $p(\mathcal{C}_k)$

$p(\boldsymbol{x})$: pdf of feature vector $\boldsymbol{x}$ ($p(\boldsymbol{x}) = \sum_{k=1}^{K} p(\boldsymbol{x}|\mathcal{C}_k)p(\mathcal{C}_k)$)

$p(\boldsymbol{x}|\mathcal{C}_k)$: pdf of feature vector $\boldsymbol{x}$ for samples of class $\mathcal{C}_k$

$p(\mathcal{C}_k)$: probability of the label be $\mathcal{C}_k$

Bayes decision rule

$K = 2$

If $P(\mathcal{C}_1|\boldsymbol{x}) > P(\mathcal{C}_2|\boldsymbol{x})$ decide $\mathcal{C}_1$
otherwise decide $\mathcal{C}_2$

$$p(error|\boldsymbol{x}) = \begin{cases} p(C_2|\boldsymbol{x}) & \text{if we decide } \mathcal{C}_1 \\ P(C_1|\boldsymbol{x}) & \text{if we decide } \mathcal{C}_2 \end{cases}$$

Bayes theorem

Bayes' theorem

Posterior Likelihood Prior

$$p(\mathcal{C}_k|\boldsymbol{x}) = \frac{p(\boldsymbol{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{p(\boldsymbol{x})}$$

Posterior probability: $p(\mathcal{C}_k|\boldsymbol{x})$

Likelihood or class conditional probability: $p(\boldsymbol{x}|\mathcal{C}_k)$

Prior probability: $p(\mathcal{C}_k)$

$p(\boldsymbol{x})$: pdf of feature vector $\boldsymbol{x}$ ($p(\boldsymbol{x}) = \sum_{k=1}^{K} p(\boldsymbol{x}|\mathcal{C}_k)p(\mathcal{C}_k)$)
$p(\boldsymbol{x}|\mathcal{C}_k)$: pdf of feature vector $\boldsymbol{x}$ for samples of class $\mathcal{C}_k$
$p(\mathcal{C}_k)$: probability of the label be $\mathcal{C}_k$

Naïve bayes

It assumes that all features are **conditionally independent given the class**

Naïve Bayes (discrete example)

Email	Contains “Free”?	Contains “Prize”?	Class
1	Yes	No	Not Spam
2	Yes	Yes	Spam
3	No	No	Not Spam
4	Yes	Yes	Spam
5	No	Yes	Not Spam
6	Yes	No	Spam
7	No	No	Spam
8	No	No	Not Spam
9	Yes	Yes	Spam
10	No	Yes	Spam
11	Yes	No	Not Spam
12	No	No	Not Spam
13	Yes	Yes	Spam
14	No	No	Not Spam
15	Yes	No	Not Spam

- Prior =
- Posterior =
- likelihood =

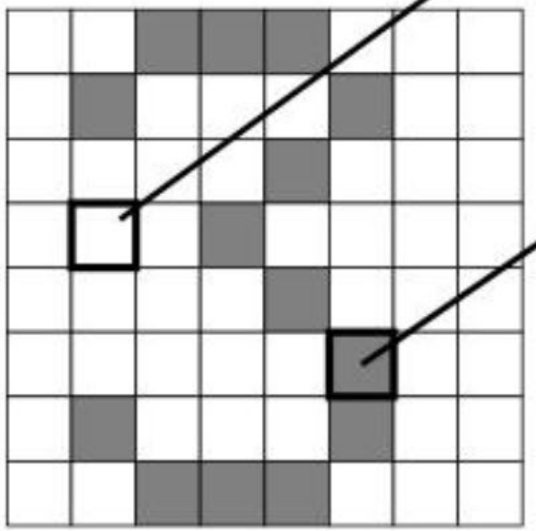
Classify an email contain the word Free

What is its probability ?

Naïve bayes- hand written digit

$P(Y)$

1	0.1
2	0.1
3	0.1
4	0.1
5	0.1
6	0.1
7	0.1
8	0.1
9	0.1
0	0.1



$P(F_{3,1} = on|Y)$

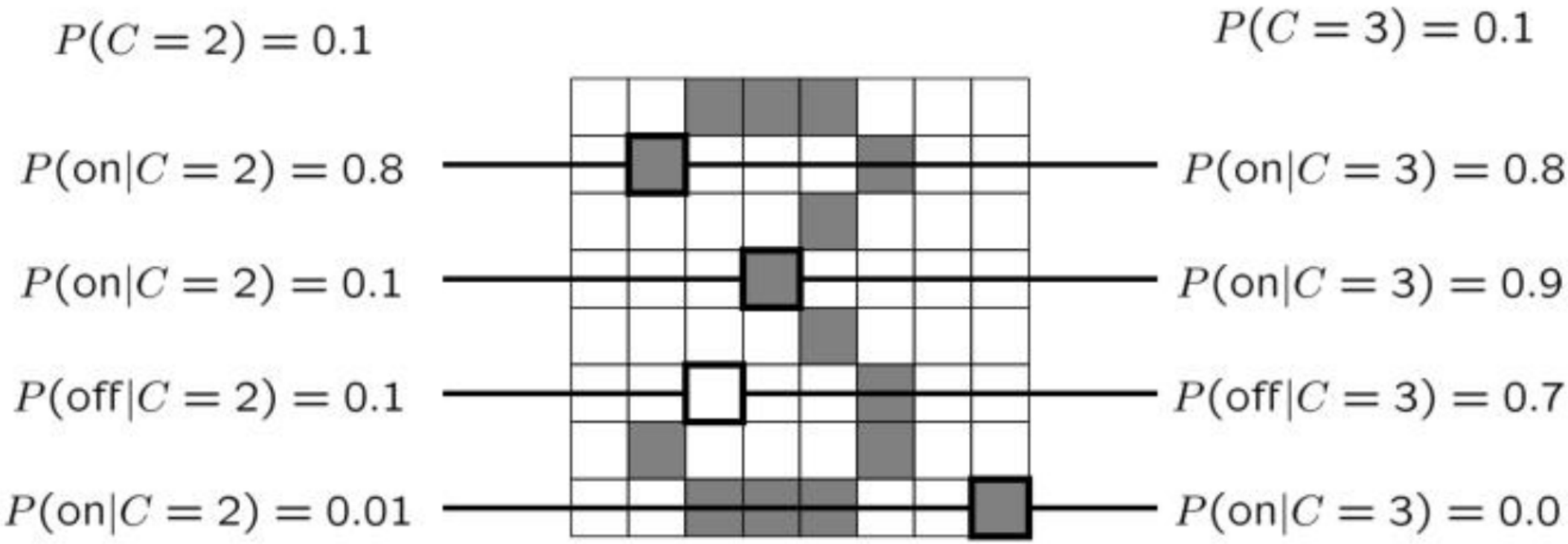
1	0.01
2	0.05
3	0.05
4	0.30
5	0.80
6	0.90
7	0.05
8	0.60
9	0.50
0	0.80

$P(F_{5,5} = on|Y)$

1	0.05
2	0.01
3	0.90
4	0.80
5	0.90
6	0.90
7	0.25
8	0.85
9	0.60
0	0.80

Problem of naïve bayes

- Overfitting example



Laplace smoothing

- How we can overcome the problem ?

- Laplace smoothing
  - Pretend you saw every outcome once more than you actually did